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Brier Score

Setting: a binary classifier outputs a probability $p_i \in [0,1]$ that $y_i = 1$. Accuracy and AUC only judge the decision or the ranking; the **Brier Score** judges the **probabilities themselves**. It is the mean squared error between predicted probability and outcome:

$$\boxed{BS = \frac{1}{N}\sum_{i=1}^{N}(p_i - y_i)^2}, \qquad y_i \in \{0,1\}, \qquad 0 \le BS \le 1, \quad \text{lower is better}$$

Example: outcomes $y = (1,0,1,0)$. Model A: $p = (0.9, 0.2, 0.7, 0.1) \Rightarrow BS = \frac{0.01+0.04+0.09+0.01}{4} = 0.0375$. Model B: $p = (0.6, 0.4, 0.6, 0.4) \Rightarrow BS = 0.16$. Both rank perfectly (AUC = 1), but A is **confident and right**.

Baseline and skill: always predicting 0.5 gives 0.25; always predicting the base rate $\bar{y}$ gives $BS_{ref} = \bar{y}(1-\bar{y})$. For rare events ($\bar{y} = 0.02$, $BS_{ref} = 0.0196$) a small BS can be misleading, so use the **Brier Skill Score**:

$$BSS = 1 - \frac{BS}{BS_{ref}} \qquad (1 : \text{perfect}, \quad 0 : \text{no better than the base rate}, \quad < 0 : \text{worse})$$

Murphy decomposition: with K bins of predictions, n_k cases, mean prediction $\bar{p}_k$ and observed frequency $\bar{o}_k$ in bin k:

$$BS \approx \underbrace{\tfrac{1}{N}\textstyle\sum_k n_k(\bar{p}_k - \bar{o}_k)^2}_{\text{Reliability: calibration error} \downarrow} - \underbrace{\tfrac{1}{N}\textstyle\sum_k n_k(\bar{o}_k - \bar{y})^2}_{\text{Resolution: discrimination} \uparrow} + \underbrace{\bar{y}(1-\bar{y})}_{\text{Uncertainty: data only}}$$

A good score needs **calibration** (when the model says 70%, it happens $\sim 70\%$ of the time) and **resolution** (predictions that separate cases instead of repeating the base rate).

Properties:

- **Strictly proper:** the expected score is minimized only by the true probability, so over- or under-confidence is always penalized.
- **vs Log Loss:** log loss explodes for confident mistakes ($p \to 0$, $y = 1$); Brier is bounded and more robust, but less sensitive between small probabilities (0.01 vs 0.001).
- **vs AUC:** AUC only measures ranking (unchanged by any monotone transform of p); Brier also penalizes miscalibration. Report both.
- **Multiclass:** $\frac{1}{N}\sum_i \sum_c (p_{ic} - y_{ic})^2$ with one-hot y, range $[0,2]$.

When it matters: whenever p is used as a number: expected loss ($p \cdot$ exposure) in credit risk, fraud thresholds, medical risk, weather forecasting (Brier, 1950), and to validate calibration (Platt, isotonic). In Python: `sklearn.metrics.brier_score_loss(y_true, y_prob)`.